AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Resilient Infrastructure Reconstruction Software

Team ID: *GEN26-A0H-I173*

Team Name: *INNOV4*

Domain: *Climate Action, Water and Environmental Resilience*

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Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
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7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals

Problem Statement

How can an explainable decision-support system help authorities compare feasible reconstruction options, assess their potential long-term consequences, and identify when changes in climate scenarios, costs or priorities could alter the preferred decision?

Problem Understanding

- **Target Users**:
  1. Government Authorities
  2. Civil Engineers
  3. Disaster Management Teams
  4. Local Communities
  5. NGOs and Relief Organizations

- **Main Causes of the Problem**:
  1. Uncertain Climate Risks
  2. Limited Budgets
  3. Poor Infrastructure Planning
  4. Incomplete Data
  5. Short-Term Reconstruction Focus
  6. Difficulty Comparing Alternatives

- **Impact on Users and Stakeholders**:
  1. Local Communities
  2. Commuters and Students
  3. Farmers and Small Businesses
  4. Government Authorities
  5. Engineers and Construction Teams
  6. Disaster-Prone Communities

Existing Gaps & Challenges

- **Limitation 1**:
  Limited Climate Risk Assessment: Existing reconstruction practices may not adequately account for future floods, landslides, and changing climate conditions.

- **Limitation 2**:
  Difficulty Comparing Reconstruction Options: Authorities may struggle to compare alternatives based on cost, long-term risks, technical feasibility, and community needs.

- **Main Challenge**:
  Developing an explainable decision-support system that compares reconstruction options and alerts authorities when changes in climate scenarios, costs, or priorities could change the preferred decision.

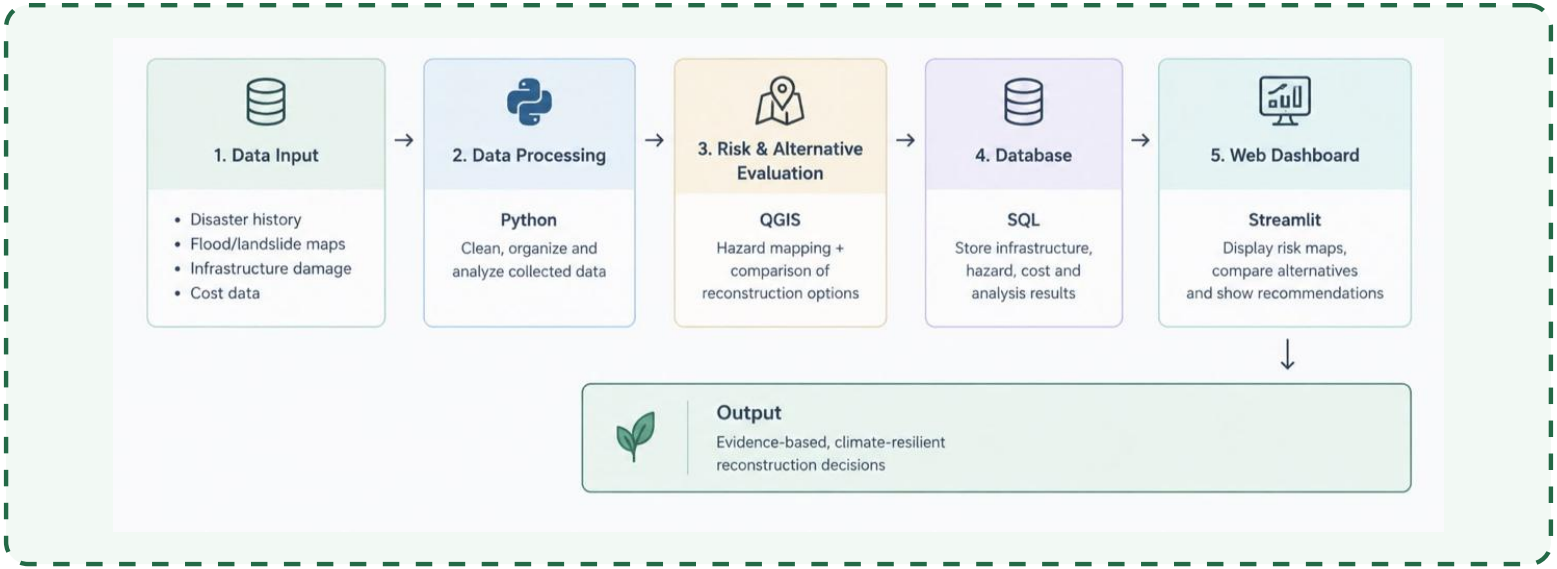
Our Proposed Solution

- **Core Idea**:
  An explainable, climate-resilient reconstruction decision-support system for roads and bridges damaged by disasters.

- **How It Works**:
  Compares reconstruction options based on cost, future climate risks, technical feasibility, and community needs. A Decision Reversal Alert highlights when changes in costs, climate scenarios, or priorities could change the preferred option.

- **Main Value**:
  Helps authorities make informed reconstruction decisions that balance immediate costs with long-term climate resilience and community needs.

Technical Approach



The system collects disaster, infrastructure and cost data to assess vulnerability. Python processes the data, while QGIS visualizes hazard-prone areas. Reconstruction alternatives are compared based on risk, cost, durability and feasibility. SQL stores the information, and a Streamlit dashboard presents the results to support informed reconstruction decisions.

Tech Stack

Frontend: Streamlit

Backend: Python

Database: SQLite

AI/ML, APIs, or Hardware: QGIS

Innovation & Uniqueness

What is new or different?

1. Integrated climate-risk and cost analysis
2. Multi-option reconstruction comparison
3. Community-needs assessment
4. Decision Reversal Alert
5. Explainable recommendations
6. Scenario-based evaluation

Why is this approach suited to the problem?

1. Future climate-risk assessment
2. Cost and feasibility comparison
3. Community-needs integration
4. Explainable decision-making
5. Decision Reversal Alert

Key Differentiator

1. Decision Reversal Alert
2. Explainable recommendations
3. Integrated risk–cost analysis
4. Community-centric planning

Feasibility & Scalability

- **Feasibility**:
  1. Python-based development
  2. Public datasets
  3. Scoring algorithms
  4. Interactive dashboard

- **Scalability**:
  1. Multi-region expansion
  2. Multiple infrastructure types
  3. Updated climate scenarios
  4. Large-scale implementation

- **Risks**:
  1. Limited data — Public datasets
  2. Climate uncertainty — Scenario analysis
  3. Biased results — Transparent scoring
  4. Technical complexity — Simple prototype

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases – Government Authorities

User: Government Authorities

Scenario:
After severe flooding damages several roads and bridges, government authorities use the system to compare reconstruction alternatives based on estimated costs, future flood risks, technical feasibility, and community needs. For example, when choosing how to rebuild a damaged bridge, the system may compare a low-cost replacement with a more flood-resilient design. It explains the trade-offs and triggers a Decision Reversal Alert if changes in the budget or climate scenarios alter the preferred option.

Expected Outcome:
Better-informed reconstruction decisions, effective budget allocation, and infrastructure that is better prepared for future climate risks.

Use Cases – Civil Engineers

User: Civil Engineers

Scenario:
After a landslide damages a mountain road, civil engineers use the system to evaluate alternatives such as rebuilding the road along its original route or choosing a feasible route with lower exposure to landslides. The system compares construction costs, technical feasibility, future hazard risks, and connectivity to nearby communities. It explains the advantages and disadvantages of each option to help engineers identify a suitable reconstruction plan.

Expected Outcome:
Safer reconstruction planning, reduced exposure to future landslides, and improved long-term road connectivity.

Use Cases – Disaster Management Teams

User: Disaster Management Teams

Scenario:
After severe flooding affects multiple villages, disaster management teams use the system to prioritize the reconstruction of damaged roads and bridges. For example, a road connecting several villages to a hospital may require higher priority than a less critical route. The system helps compare infrastructure based on community accessibility, urgency, reconstruction cost, and future climate risks, supporting the allocation of limited recovery resources.

Expected Outcome:
Prioritized infrastructure recovery, improved access to essential services, and more effective disaster recovery planning.

Impacts and Benefits

- **Benefit to Target Users**:
  Helps government authorities, civil engineers, and disaster management teams compare reconstruction options and make informed, climate-resilient decisions.

- **Social, Economic, and Environmental Impact**:
  Improves community safety and accessibility, reduces potential future reconstruction costs, and encourages sustainable infrastructure planning.

- **Expected Measurable Outcome**:
  Reduced exposure to climate risks, improved infrastructure prioritization, more efficient budget allocation, and better access to essential services.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Promotes resilient infrastructure and sustainable development.
SDG 13	Target 13.1	Climate Action	Strengthens resilience and adaptive capacity to climate-related hazards.

Individual Member Contribution

Member	Contribution or responsibility
Danushree K S	Team Leader. Collecting data from resources
Nibashini V G	Verifying data accumulated
Kavyashree Rupesh	Handled technical aspects like editing and ppt creation
Shamritha D	Designing chartwork

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